

Anurag Thakur

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Generative AI, RAG pipelines, and multi-agent systems using LangChain, CrewAI, and vector databases. Skilled in Python, prompt engineering, and workflow automation (n8n, Zapier), with a focus on production-ready, enterprise-grade AI solutions.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS, C

Generative AI: LLMs, RAG, LangChain, CrewAI, Langsmith, Prompt Engineering, NLP

Vector Databases: Databases: FAISS, Pinecone, ChromaDB, Qdrant, PostgreSQL, Firebase

ML & DL: Scikit-Learn, TensorFlow, PyTorch, OpenCV, Pandas, NumPy, PowerBI, Excel

Backend & APIs: Flask, Streamlit, RESTful APIs, FastAPI

Workflow Automation: n8n, Zapier

Cloud & DevOps: AWS (EC2, S3), Docker, Git, CI/CD Pipelines, DVC

EXPERIENCE

AI/ML Engineer Intern

Nov 2025 – May 2026

Coder Roots

- Designed and deployed 4+ production-grade ML pipelines for classification, regression, and clustering tasks using Scikit-Learn and XGBoost, applying rigorous evaluation frameworks and performance optimization.
- Built and deployed RESTful APIs for ML inference services with comprehensive documentation, enabling seamless integration with downstream systems and real-time predictions.
- Trained CNN and transfer-learning models using TensorFlow and PyTorch, achieving robust validation accuracy while managing multiple project priorities independently.

PROJECTS

GitHub Code Explainer (AI Agent) 🤖🌐

- Architected an autonomous AI agent that indexes GitHub repositories and answers developer queries via RAG, demonstrating expertise in agentic AI design and tool-using agents.
- Integrated Pinecone vector database, LangChain, and Groq Llama 3 for semantic code search, achieving sub-2s query latency with optimized retrieval pipelines.
- Designed REST endpoints for repository indexing and concurrent multi-repo session management, supporting real-time developer interactions.

PDF Q&A Chatbot (RAG) 🤖

- Built a production-ready RAG chatbot for querying user-uploaded PDFs, eliminating manual document lookup and demonstrating expertise in unstructured data processing.
- Implemented chunking strategies, FAISS vector indexing, and semantic retrieval using Hugging Face embeddings to accelerate document search and retrieval.
- Generated context-grounded responses with LangChain, constraining outputs to retrieved content to eliminate hallucinations and ensure reliable answers.

Energy Consumption Forecasting 🤖🌐

- Developed an XGBoost-based forecasting application on 121,000+ hourly records from PJM grid data, demonstrating strong data analysis and feature engineering skills.
- Engineered time-based features (lag windows, hour, day, seasonality) improving R^2 from 0.45 to 0.87 through iterative optimization.
- Deployed an interactive Streamlit dashboard with date-range selection, peak-consumption metrics, and trend-analysis charts for real-time data exploration.

Driver Drowsiness Alert System 🤖

- Built a real-time drowsiness detection system using OpenCV and dlib for live video feed processing with sub-2s alert latency.
- Applied Eye Aspect Ratio (EAR) thresholding to distinguish prolonged eye closure, implementing guardrails to minimize false positives.

EDUCATION

D.A.V University

Bachelor of Technology in Computer Science & Engineering

- CGPA: 6.57
- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures, Algorithms
- Participated in an inter-college technical hackathon (CGC Jhanjeri, Nov 2023), building a working prototype within a 24-hour constraint, demonstrating curiosity, initiative, and a commitment to learning emerging technologies.